

Methane Fluxes Revisited: A Coupled Perspective

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Abstract

This paper reports a systematic evaluation of drought-sensitive methane fluxes. We release the processing scripts and intermediate artefacts to support independent verification. The reduced model retains most of the explanatory power at a fraction of the computational cost. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

1. Introduction

The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting.

Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification.

Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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2. Related Work

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3. Materials and Methods

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4. Experimental Setup

Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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5. Results

Residual error grows approximately linearly with the number of coupled parameters. Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. Sensitivity to the smoothing window proved modest across the range we examined.

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6. Discussion

We release the processing scripts and intermediate artefacts to support independent verification. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Taken together, these findings argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol.

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